

Smart-City Mobility: Traffic Intelligence Insights Report

Based on Metro Interstate Traffic Volume Dashboard (2015–2017) Prepared for:
Smart-City Mobility & Transport Planning Team

1. Introduction

This report summarises key insights derived from the Power BI Traffic Intelligence Dashboard built using historical interstate traffic data (2015–2017). The analysis focuses on temporal patterns, weather impacts, and temperature-traffic relationships to support data-driven mobility planning, congestion mitigation, and operational decision-making for smart-city initiatives.

2. Data Quality & Preparation Summary

Before analysis, the dataset was cleaned and enhanced using Power Query:

- Verified **row and column counts** and ensured structural completeness.
- Identified missing values (mainly in *weather_description*), with no major impact on core metrics.
- Corrected data types for *date_time*, *traffic_volume*, *temp*, and weather fields.
- Engineered new fields:
 - **Hour** extracted from DateTime
 - **Temperature in Celsius (Temp_C)**
 - **Traffic Category** (Low, Medium, High) based on volume thresholds

These transformations enabled more granular temporal and environmental analysis.

3. Key Insights

3.1 Daily Traffic Trends (2015–2017)

Daily average traffic patterns show clear and consistent commuter behaviour:

- **2015:** Lower volumes due to partial-year data.
- **2016:** Strong weekday peaks and weekend troughs, indicating predictable commuter flows.
- **2017:** Highest overall traffic, with more pronounced morning and evening peaks.

Implication for mobility planning: The city can anticipate recurring congestion windows and optimise signal timing, ramp metering, and public transport scheduling around morning (7–9 AM) and evening (4–6 PM) peak periods.

3.2 Hourly Traffic Patterns (2017)

Hourly analysis reveals:

- **Low traffic:** 12 AM–5 AM
- **Morning peak:** 7–9 AM
- **Midday dip:** 11 AM–2 PM
- **Evening peak:** 4–6 PM

Implication: These patterns support targeted deployment of traffic enforcement, dynamic lane management, and real-time traveller information systems during peak hours.

3.3 Weather Impact on Traffic

Average traffic volume varies significantly across weather conditions:

- **Highest traffic:** Typically during *Clear* or *Cloudy* conditions
- **Lowest traffic:** Often during *Snow*, *Fog*, or *Mist*
- **Difference:** A substantial drop in volume between best and worst conditions (exact values depend on dataset)

Implication: Adverse weather reduces mobility efficiency. Smart-city systems should integrate weather-responsive traffic control, including adaptive speed limits, hazard alerts, and increased public transport frequency during low-visibility conditions.

3.4 Temperature vs Traffic Volume

Scatter plot analysis shows:

- **Weak positive relationship:** Traffic slightly increases with moderate temperatures.
- **Higher traffic range:** Typically between **10°C–20°C**, where commuter activity is highest.
- **Outliers:**
 - High traffic at low temperatures (possible holiday events or incidents)
 - Low traffic at normal temperatures (likely night-time hours)

Implication: Temperature alone is not a strong predictor of traffic, but extreme conditions may require operational adjustments such as increased monitoring or targeted messaging to drivers.

4. KPI Summary

Three KPIs were created to support high-level monitoring:

- **Total Hours Analysed** – Indicates dataset coverage and reliability.
- **Average Traffic Volume** – Provides a baseline for congestion assessment.
- **Average Temperature** – Supports environmental correlation analysis.

These KPIs help mobility planners quickly assess system-wide conditions and compare trends across years.

5. Strategic Implications for Smart-City Mobility

Based on the dashboard insights, the following actions are recommended:

1. Peak-Hour Congestion Management

- Deploy adaptive traffic signal control during morning and evening peaks.
- Encourage modal shift through targeted public transport incentives.

2. Weather-Responsive Traffic Operations

- Integrate weather data into traffic prediction models.
- Activate dynamic speed limits and hazard warnings during fog, snow, or heavy rain.

3. Infrastructure Planning

- Prioritise road maintenance and visibility improvements in areas prone to weather-related slowdowns.
- Consider expanding capacity or alternative routes for high-volume corridors.

4. Smart Mobility Services

- Provide real-time congestion alerts through mobile apps and roadside signage.
- Use hourly and daily patterns to optimise ride-sharing, bus frequency, and autonomous vehicle routing.

6. Conclusion

The Traffic Intelligence Dashboard provides a robust foundation for understanding mobility patterns across time, weather, and environmental conditions. These insights enable the smart-city mobility team to make informed decisions that enhance traffic flow, improve commuter experience, and support long-term urban mobility planning.

With continued integration of real-time data, predictive analytics, and IoT sensors, the city can evolve toward a fully adaptive, intelligent transportation ecosystem.